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Learning Tonotactic Patterns with Autosegmental Representations

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Abstract

This paper presents and evaluates an algorithm which learns tonotactic patterns over autosegmental representations (ARs), facilitated in part by showing how adjacency matrix representations can represent ARs. The Bottom-Up Factor Inference Algorithm over ARs (BUFIA-AR) takes AR graphs as input, and outputs a set of inviolable subgraphs as constraints. The input AR graphs themselves are generated automatically from linear transcriptions. A corpus of 621 non-derived native Hausa words was used to test BUFIA-AR, which returned 11 constraints. The results fall into three categories: constraints that align with previous generalisations, constraints that provide more specific generalisations than previously posited and which better account for the data, and constraints that identify novel patterns. These results demonstrate the feasibility and efficiency of ARs as a computational representation for phonotactic learning.

Contents

1. Introduction	1
2. Learning Tonotactics	3
2.1 Tonal Mapping Problem	3
2.2 Learning Models	6
3. Preliminaries	7
3.1 Strings Representations	7
3.2 Autosegmental Representation as Graphs	8
3.3 Structure Containment	10

1. Introduction

This paper provides one method for learning tonotactic patterns over autosegmental representations (ARs). A key insight of phonological theory is to view forms in tonal languages as *autosegmental graphs*, rather than one-dimensional linear strings (Goldsmith, 1976; Coleman and Local, 1991; Jardine, 2017; Oakden, 2020). These

representations have been argued to have demonstrated advantages for analysing tonal patterns (Hyman, 2009; Clements and Goldsmith, 2010; Odden, 2013). Computationally, grammars over ARs have been shown to express both local and long-distance constraints, while maintaining a relatively low level of computational complexity (Jardine, 2017; Chandlee and Jardine, 2019). However, despite the importance of ARs in tonal phonology and previous research on their formal properties (Kornai, 2007), there has been limited work on learning tonal grammars, and no phonotactic learning models to our knowledge have adopted ARs.

This paper focuses on learning tonotactic grammars. Following Jardine (2017), the tonotactic grammar consists of a set of language-specific inviolable constraints over ARs. The constraints themselves only specify marked AR structures. Logically, such grammars are simply the conjunction of negative literals (Rogers and Lambert, 2019). In this setting, the tonotactic learning problem boils down to identifying those constraints (the marked AR structures) which are consistent with the entire dataset.

The Bottom-Up Factor Inference Algorithm (BUFIA) is a learning algorithm that deterministically searches a space of possible constraints and is provably guaranteed to find the most general constraints consistent with its input data (Chandlee et al., 2019; Rawski, 2021). Crucially, it is couched in model theory (Rogers, 1998; Enderton, 2001) so that it can operate with any particular representational scheme.

This work presents BUFIA-AR, an instantiation of BUFIA that learns tonotactic constraints expressed with ARs. As will be explained, this implementation utilises a matrix-based representation for ARs, which is an adaptation of matrix-based representations of graphs (West et al., 2001; Cormen et al., 2022).

BUFIA-AR was tested on a curated corpus of Hausa (Awagana et al., 2009), a Chadic language in West Africa. The corpus consists of 621 mono-morphemic words, which altogether provides a collection of 64 distinct ARs. Using this corpus as input, we assess the performance of BUFIA-AR through a combination of qualitative and quantitative evaluations. Qualitatively, BUFIA-AR returns a grammar consisting of 11 AR constraints when syllable, tone, and mora counts are limited to three. These BUFIA-AR-learned constraints either align with, or are more specific than, those previously proposed in the literature. The algorithm additionally identifies novel patterns that have not been previously reported. Quantitatively, a cross-validation experiment shows that BUFIA-AR achieves a mean accuracy over 90% in both token- and type-based settings which outperforms the linguistic baseline grammar and a BUFIA learner using string representations (BUFIA-Str). Altogether, these results demonstrate the feasibility of using BUFIA-AR for tonotactic grammar inference. All data and programs used in this paper are accessible on GitHub.

In this way, this paper takes a first step towards learning tonal grammars expressed in terms of ARs. We acknowledge that tonal grammars are much more than just the tonotactic component. The grammars that BUFIA-AR returns have nothing to say about how underlying representations of tonal languages are learned, nor how these underlying representations are transformed into surface representations. We also acknowledge that some surface constraints have been argued to follow from the tonal processes themselves in some languages (see Section 2), which is again something the grammars returned by BUFIA-AR cannot express. Nonetheless, despite these limitations,

BUFIA-AR still makes a contribution to our understanding of how tonal grammars expressed in terms of ARs can be learned. We expect that some of the insights made here will persist in future research which addresses these broader problems regarding the learning of tonal grammars.

This paper is structured as follows. Section 2 provides an overview of the tonotactic learning problem and current models. Section 3 introduces formal language theory, model theory, and graph theory relevant to ARs and BUFIA. Section ?? provides a detailed description of the BUFIA-AR implementation. Section ?? presents the Hausa case study and model evaluation. Section ?? discusses issues relating to constraint optimisation, constraint satisfaction, and constraint weighting. Section ?? concludes the paper and points to some future directions.

2. Learning Tonotactics

2.1. Tonal Mapping Problem

A central question in tonal phonology concerns how languages organise tones, syllables, and/or moras into well-formed structures. A closely related modelling question asks what kinds of well-formed tonal patterns are attested in a language, and how such patterns can be systematically characterised. To illustrate these issues, we begin with a frequently discussed example: the tonal mapping problem in two West African tonal languages, Hausa and Mende. Hausa is a Chadic language, while Mende is a South-western Mande language spoken in Sierra Leone (Leben, 1978). Both languages use High (H) and Low (L) tones as basic tonemes with the mora as the tone bearing unit (TBU). However, they show distinct patterns in how their tonal melodies are mapped to TBUs (Newman, 2002; Leben, 1978; Zoll, 2003; Jardine, 2017).

As shown in Table 1, the two languages first differ in their monosyllabic contour inventory. Mende permits a wider range of monosyllabic contours, including light HL, heavy HL, and heavy LH, whereas Hausa allows only the heavy HL contour. Secondly, when a two-tone melody maps onto a trisyllabic word, Mende and Hausa have distinct surface forms: HL surfaces as H.L.L in Mende (*fě.là.là* ‘junction’) but as H.H.L in Hausa (*jée.máa.gèe* ‘bat’); similarly, LH surfaces as L.L.H in Mende (*lè.lè.má* ‘mantis’) but as L.H.H in Hausa (*jì.mí.ná* ‘ostrich’)^{1,2}.

There are mainly three approaches to the tone mapping problem. The first identifies three key factors that determine the surface forms: *directionality* (whether tone spreads rightwards or leftwards), *morphological categories* (prefix or suffix), and *tone quality* (whether spreading is sensitive to H or L) (Hyman, 1987; Leben, 2017; Hyman, 2009).

¹This paper follows the tonal transcription method commonly used by Africanists. For instance, “H.H.L” represents two H-toned syllables followed by a L-toned syllable. In addition, tones are always marked on the nuclear vowels. Long vowels are transcribed as double vowels.

²Dwyer (1978) offers a different analysis of Mende tones than Leben (1978)’s, and provides supplemental data in which L.H.H forms also occur (e.g. *ndávúlá* ‘sling’). Leben (1978) account for the different patterns by proposing two distinct mappings.

	Mende			Hausa	
contour inventory	$\begin{array}{c} \text{H L} \\ \diagdown \diagup \\ \text{mbu} \\ \text{'owl'} $	$\begin{array}{c} \text{L H L} \\ \quad \diagdown \diagup \\ \text{ngewoo} \\ \text{'God'} $	$\begin{array}{c} \text{L H} \\ \diagdown \diagup \\ \text{mbaa} \\ \text{'rice'} $	$\begin{array}{c} \text{H L} \\ \diagdown \diagup \\ \text{mai} \\ \text{'oil'} $	
tone mapping	$\begin{array}{c} \text{H L} \\ \quad \diagdown \diagup \\ \text{f e l a l a} \\ \text{'junction'} $	$\begin{array}{c} \text{L H} \\ \diagdown \diagup \\ \text{l e l e m a} \\ \text{'mantis'} $	$\begin{array}{c} \text{H L} \\ \diagdown \diagup \\ \text{j e e m a a g e e} \\ \text{'bat'} $	$\begin{array}{c} \text{L H} \\ \quad \diagdown \diagup \\ \text{j i m i n a} \\ \text{'ostrich'} $	

Table 1: Contour inventories and tone mapping in Mende and Hausa.

The second account is Optimal Tone Mapping (OTM; [Zoll, 2003](#)). Zoll argues that directionality-based accounts of phonological mapping can in some cases yield incorrect surface predictions. Under OTM, surface tonal patterns emerge from language-specific rankings between morphological faithfulness constraints and markedness constraints, CLASH and LAPSE, which are sensitive to tone quality.

The third approach, proposed by [Jardine \(2017\)](#), is to use language-specific, inviolable constraints, where the constraints are marked, forbidden ARs; that is, ARs to be avoided. For example, the No SHORT RISE constraint in Mende ([Leben, 2017](#)) that bans monomoraic LH can be represented as a banned AR as in Figure 1a.

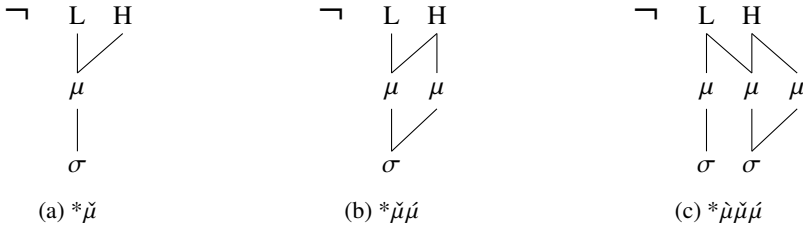


Figure 1: The forbidden structure in (a), $*\check{\mu}$ (corresponding to No SHORT RISE), and two ARs in (b–c) containing it.

Following [Jardine \(2017\)](#), if Figure 1a is forbidden in the language, any AR *containing* 1a, such as 1b and 1c, will also be forbidden. Formally, if Grammar G is the set of constraints in the language, such that $G = \{AR_1, AR_2, \dots, AR_n\}$, a well-formed We represent factors as *conjunctive negative literals*: $\neg AR_1 \wedge \neg AR_2 \wedge \dots \wedge \neg AR_n$ ([Jardine, 2017](#); [Rogers and Lambert, 2019](#)). It follows that the tonotactic learning problem is to identify a set G of banned ARs given a set of example surface forms. This paper follows the third approach of [Jardine \(2017\)](#).

One potential criticism of [Jardine \(2017\)](#), anticipated by the author and also applicable to the present work, is that it explains tonal patterns exclusively in terms of

surface-true constraints. One consequence is that well-formed patterns that are not directly observable in surface forms, or at the lexical level, are not realised by the grammar. The grammar inferred at this level therefore reflects only the final outcomes of tonal mapping, rather than intermediate representations produced by tonal rules, which potentially misses important phonological generalisations. **[Do we want to mention ICGI paper here - maybe we should, why not]**

Hausa provides a clear illustration of this point. The absence of rising tones on a single syllable can be explained as follows. An underlying LH sequence may arise as an intermediate representation, either through synchronic phonological processes or as the result of historical change, but it is subsequently simplified to either H in non-final position or L elsewhere (Leben, 1971; Newman, 2002). Likewise, the absence of final L.L sequences on long vowels is attributed to LOW TONE RAISING, which changes underlying L.L to L.H when the second L occurs on a long vowel (Leben, 1971, 1996). **[I removed the SPE styled rules and explained in prose instead.]**

This example illustrates that unattested surface patterns do not necessarily reflect illicit structures in the grammar itself, but rather intermediate representations that are subsequently subject to some repair processes. While acknowledging the limitations of these grammars, we argue that learning such grammars is nonetheless valuable. They still provide meaningful insights into tonotactic patterns and serve as a foundation for further work for three reasons.

First, a grammar containing a set of forbidden ARs that is consistent with observable surface forms is not in conflict with a grammar for the tonal processes; rather, it captures their pronounceable outcomes. Given available access to a lexicon at a different level of abstractness, another grammar could likewise be learned, and these two grammars can be systematically compared. This comparison between different levels of abstractness itself yields non-trivial insights into tonal processes.

Second, considering that ARs have been long used in tonal phonology, yet absent from computational learning, we believe a more pressing priority is establishing representational adequacy for learning tones, which is the goal of this work. Without an explicit framework that encodes the geometry of tonal representations, modelling more complex tonal alternations, such as those discussed above, is more difficult. Establishing this representational foundation is therefore a useful precondition for future work on deeper tonal processes. **[On a second reading, I think this paragraph is still technically correct, but it now feels somewhat outdated. It reads like that we dont know whether transformations can be modeled over ARs, but we now know its possible, and exactly how do to]**

Lastly, we are not aware of existing learning models that operate with ARs directly for either tonal processes or tonotactics. In the next section, we review previous learning approaches related to tier-based representations and learning, and then introduce BUFIA, the algorithm that our model is based on.

2.2. Learning Models

This section reviews systems for phonotactic and tonotactic learning. One line of research employs statistical methods from machine learning (Coleman and Pierrehumbert, 1997; Hayes and Wilson, 2008; Goldsmith and Riggle, 2012). These methods have been deployed on string and syllabic representations for learning segmental phonotactic patterns as well as stress patterns, but have not been used for learning tonotactics with ARs.

Gouskova and Gallagher (2020) synthesises the Maximum Entropy (MaxEnt) strategy of Hayes and Wilson (2008) with a procedure that induces phonological tiers in order to learn non-local phonotactics. Related work by Jardine and Heinz (2016), Jardine and McMullin (2017), De Santo and Aks nova (2021), and Lambert (2021), and Swanson (2025) provide theoretical results and algorithms regarding the induction of phonological tiers for non-local phonotactics. Other work has proposed procedures for learning tiers for grammars modelling non-local phonological processes (Burness and McMullin, 2019, 2021; Belth, 2024). While the successful induction of tiers from data is an important achievement of the 21st century, the constraints induced over the projected tiers are still strings, and not ARs. For example, none of the aforementioned methods can learn a constraint that describes a many-to-one association from the tonal tier to the TBU tier or vice versa.

BUFIA represents another approach (Chandlee et al., 2019; Rawski, 2021). BUFIA differs from the aforementioned approaches in a couple of ways. First, it is not specific to any particular representation or data structure, such as strings or trees. For this reason, Chandlee et al.’s (2019) presentation of BUFIA abstracts away from a specific representation, and presents BUFIA in terms of arbitrary model-theoretic relational structures. As long as the representations of interest can be expressed model-theoretically, BUFIA can be implemented. Since strings, trees, and graphs are all examples of structures that have model-theoretic representations, versions of BUFIA can in principle be designed for finding constraints over any of these representations. This generality not only allows it to be instantiated and ultimately deployed for the learning of a variety of constraint types, but it also makes clear how the structure of the space of possible constraints facilitates learning.

Second, BUFIA is a deterministic learning algorithm and in its current form uses no statistical methods or randomness in its decision-making. Basically, BUFIA checks possible constraints against the data one at a time, adding those that are consistent with the data. It returns a grammar consisting of inviolable constraints. As such, this grammar is a conjunction of negative literals: $\neg C_1 \wedge \neg C_2 \wedge \dots \wedge \neg C_n$, where each C_i is a constraint, that is, a forbidden relational structure.

As will be explained in Section ??, BUFIA works because any representational scheme orders the expressible constraints in a natural way. By beginning with the smallest constraint, BUFIA is guaranteed to search as much of the space as possible and return the most general constraints (that is, the smallest forbidden structures) consistent with the data, before a stopping condition is met. Furthermore, BUFIA has been shown to be at least as successful in learning feature-based, segmental phonotactics of Quechua (Swanson et al., 2026) as phonotactic learning models based on the principle of Maximum Entropy (Hayes and Wilson, 2008; Wilson and Gallagher, 2018).

However, BUFIA has not previously been used in tonotactic learning over non-string representations.

To sum up, while ARs have long been conceptualised as graph structures, no existing work learns grammars with ARs, nor has BUFIA been instantiated to operate with ARs. To address this gap, we present BUFIA-AR, a BUFIA-based learner capable of learning tonotactic constraints directly over ARs. To our knowledge, BUFIA-AR is the first learning algorithm that directly infers such constraints from ARs. As an instantiation of the original BUFIA framework, BUFIA-AR inherits its strengths and limitations.

A more detailed presentation of BUFIA-AR is given in Section ?? . Before turning to that discussion, we first introduce in Section 3 several elements from formal language theory, model theory, and graph theory that are essential for understanding the algorithm and its internal structure. Some of this background knowledge is well established in computer science (Cormen et al., 2022; West et al., 2001); our presentation is adapted to suit those only familiar with phonological theory.

[I added the paragraph of math disclaimer back but Im just worried that this is what R3 asked us to remove...]

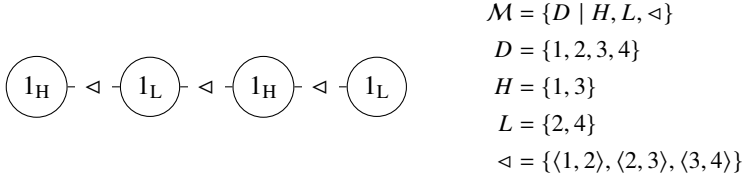
We note that Section 3 and ?? are somewhat technical, and that researchers only interested in BUFIA-AR’s performance on the Hausa dataset may find them cumbersome. However, we believe these sections are important to include because they provide the necessary details for someone (1) to understand how BUFIA-AR is an instantiation of the more general BUFIA algorithm, (2) to implement the algorithm themselves (in order to replicate the study we conducted), and (3) to understand how the algorithm’s behaviour as well as how it reflects the geometry of tonal phonology.

3. Preliminaries

In this section, we begin with structures for representing strings (Section 3.1), and then extend them to ARs using graphs and matrices (Section 3.2). Central to BUFIA is Model Theory (Enderton, 2001), which views structures as mathematical objects, and uses formal logic to represent structural units and the relationships between them. Model-theoretic analysis has been used to compare differences between different representational theories of phonological structure (Strother-Garcia, 2019; Oakden, 2020; Jardine et al., 2021). The matrix-based representation used here is a common method in mathematics and a particularly convenient way to represent AR graphs. These formal representations are essential for subsequent algorithm development, thereby understanding of the properties and learnability of ARs.

3.1. Strings Representations

Tonal melodies can be represented as strings, that is, sequences of symbols. Let Σ be a finite alphabet of symbols and Σ^* the set of all strings over Σ . For example, in a language with a tonal inventory $\{H, L\}$, we have $\Sigma = \{H, L\}$, and Σ^* contains all possible strings over Σ , such as HL , HHL , $HLHL$, and so on. Let $|w|$ denote the length of a string w . For $0 \leq n < m \leq |w|$, let $w[n : m]$ denote a non-empty substring of w .

Figure 2: String model for *HLHL* over $\Sigma = \{H, L\}$.

starting at position n and ending just before position m . The empty string ϵ is trivially a substring of every string.

A string can also be represented *model-theoretically* with two components: the domain D and a finite set of n -ary relations $\mathcal{R} = \{R_1, R_2, \dots, R_i\}$ (Rogers et al., 2013; Chandlee et al., 2019; Rogers and Lambert, 2019). A string model of *HLHL* is shown in Figure 2. In this model, $D = \{1, 2, 3, 4\}$ indicates that there are four elements in the string. Positions $\{1, 3\}$ are labelled with the unary relation H , while $\{2, 4\}$ are the positions labelled with L . Elements are connected via a binary *successor* relation, denoted by $\triangleleft$. For example, $\triangleleft = \{\langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle\}$ specifies that position 1 (labelled H) immediately precedes position 2 (labelled L), position 2 immediately precedes position 3, and so on.

3.2. Autosegmental Representation as Graphs

Unlike the string model, where elements are ordered linearly along a single axis, an AR distributes linguistic information across multiple tiers. In this work, in addition to the basic tone and syllable tiers, we incorporate a mora tier, and treat the mora as a secondary unit subordinated to syllables to indicate the syllable weight. Therefore, we assume each syllable must associate with at least one mora. Figure 3a illustrates a disyllabic form with an LH.H melody, which is visualised as a graph in Figure 3b, which corresponds to the model-theoretic relational structure in Figure 3c.

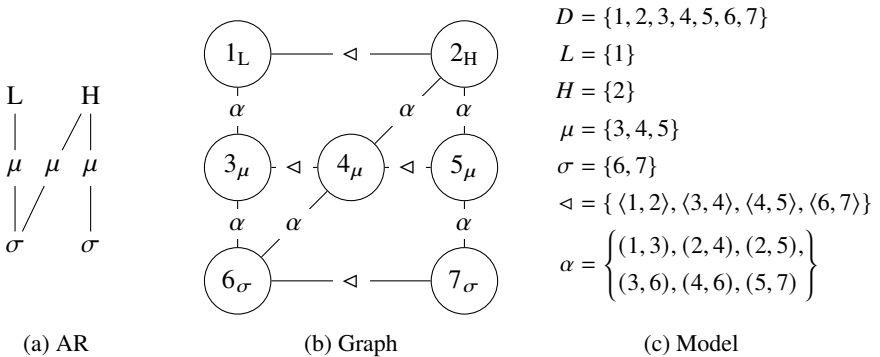


Figure 3: Three representations of a disyllabic LH.H

A graph consists of a set of vertices V and edges E . As can be seen in Figure 3c, the graph consists of seven *labelled nodes* which are circles numbered with $\{1, 2, \dots, 7\}$, representing linguistic units, each labelled with its corresponding properties, such as L , H , σ , or μ . The lines connecting elements are referred to as *edges*, which may be either *directed* or *undirected*. For clarity, we distinguish two types of connections: *association relations*, denoted by α , refer to the undirected edges across tiers; and *successor relations*, denoted by $\triangleleft$, represent ordered relationships within a single tier. All the successor connections in the current example are given by $\triangleleft = \{\langle 1, 2 \rangle, \langle 3, 4 \rangle, \langle 4, 5 \rangle, \langle 6, 7 \rangle\}$. The association relations, on the other hand, are expressed as a set of unordered pairs: $\alpha = \{(1, 3), (2, 4), (2, 5), (3, 6), (4, 6), (5, 7)\}$. Together, the tuple $\langle H, L, \mu, \sigma, \triangleleft, \alpha \rangle$ constitutes the *signature* of this model-theoretic representation of an AR.

In addition to representing model-theoretic relational structures with graphs, one can use *adjacency matrices* (West et al., 2001; Cormen et al., 2022), which use binary values to denote whether there is an edge between two nodes. Two nodes, v and w , are considered *adjacent* if there is an edge connecting them. If G is a graph with n nodes then its adjacency matrix A is an $n \times n$ matrix. If i and j are connected, the entry (i, j) has the value of 1; otherwise 0. Using this method, an AR with t tones, m moras, and s syllables can be encoded as a $t \times m$ tone-mora matrix and a $m \times s$ mora-syllable matrix, with successor relations implicit in the indexing.

For instance, Table 2 provides alternative representation of the same AR in Figure 3. In the tone-mora matrix, the header row reflects the tonal sequence LH , and the index column reflects the mora sequence. Since the L is associated with the first mora μ_1 , and the H is associated with both μ_2 and μ_3 , the corresponding cells (L, μ_1) , (H, μ_2) , (H, μ_3) are marked with 1, and others with 0 which indicates no connection between them.³ If we examine the matrix by columns, one can interpret that the L tone occupies the first mora, while the H tone occupies the remaining two. Likewise, in the mora syllable matrix, since σ_1 is associated with μ_1 and μ_2 , and σ_2 with μ_3 , the corresponding cells are marked with 1. If we examine this matrix by rows, it can be seen that σ_1 is heavy while σ_2 is light. Hereafter, we refer to the rows and columns in the tone-mora matrix as the mora rows and tone columns respectively; and to the rows and columns in the mora-syllable matrix as the syllable rows and mora column respectively.

	L	H			
μ_1	1	0	σ_1	1	1
μ_2	0	1	σ_2	0	0
μ_3	0	1			1

Table 2: Adjacency matrices: tone–mora matrix (left) and mora–syllable matrix (right)

³For convenience, we refer to each cell by the labels of the nodes. Node indices will be omitted from here on. For example, we may refer to $(1, 3)$ and $(2, 4)$ with (L, μ_1) and (H, μ_2) , respectively. However, these are the labels of the nodes rather than the nodes themselves.

These AR adjacency matrices capture precisely the same information as the graphs and model-theoretic structures. If we replace the header row and index column with the node numbers used in the graph model signature, and replace the 1s with the indices of row and column, we obtain the table filled as in Table 3. As can be seen, these pairs are identical to the information encoded in Figure 2 **[It might be more consistent/clear to replace the $\rightarrow$ with $\triangleleft$, as in the one on the left]**

	1	$\triangleleft$	2		3	$\rightarrow$	4	$\rightarrow$	5
3	(1, 3)		0	6	(3, 6)		(4, 6)		
Δ				$\downarrow$					
4	0		(2, 4)	7					(5, 7)
Δ									
5	0		(2, 5)						

Table 3: Equivalent node-indexed encodings of the adjacency matrices in Table 2, where $\triangleleft$ and α are the successor and association predicates of the model signature.

Furthermore, certain patterns can be easily identified by inspecting the matrices. For example, a series of contiguous 1s may indicate tone spreading over multiple moras if observed in the tone column, or a mora shared by multiple tones if seen in the mora row. Two 1s positioned diagonally suggest a one-to-one mapping. The associations (L, μ_1) and (H, μ_2) in the tone-mora matrix in Table 2 present an example of this. Some particular or ill-formed patterns can also be readily detected, which will be discussed in Section ??.

3.3. Structure Containment

An important concept when considering model-theoretic relational structures, either strings, sequences of feature bundles, or ARs, is *containment*. Intuitively, containment refers to the situation when one structure includes another structure inside of it. We formalise this notion using two related definitions, following [Chandlee et al. \(2019\)](#): *restriction* (Definition 3.1) and *containment* (Definition 3.2). Readers are referred to [Rogers and Lambert \(2019\)](#) for an alternative formulation. In the definitions below, it is assumed the structures A and B have the same signature $\langle R_1, \dots, R_n \rangle$.

Definition 3.1. $A = \langle D^A; R_1^A, \dots, R_n^A \rangle$ is a restriction of $B = \langle D^B; R_1^B, \dots, R_n^B \rangle$ iff $D^A \subseteq D^B$ and for each m -ary relation R_i^A , we have $R_i^A = \{(x_1, \dots, x_m) \in R_i^B \mid x_1, \dots, x_m \in D^A\}$.

Definition 3.2. Structure A is contained in structure B ($A \sqsubseteq B$) if there exists a restriction of B, denoted B' , and there exists a function $h : A \rightarrow B'$ such that for each m -ary relation R_i in the model signature: $R_i(a_1, \dots, a_m) \in A \implies R(h(a_1), \dots, h(a_m)) \in B'$.

These definitions mean that A is *contained* in B whenever there exists a subpart of B onto which A projects via a function h , and furthermore, that the relations that hold among elements of A hold among their corresponding elements in B. For instance, the AR in Figure (??) is contained in the AR in (??) because the dotted region in

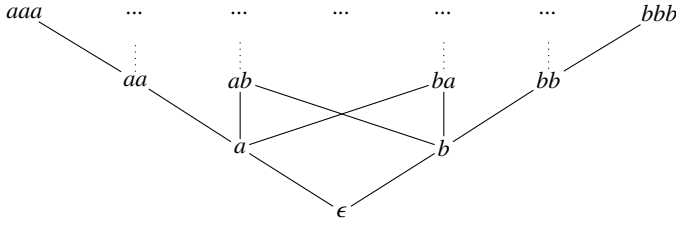


Figure 4: Strings over $\Sigma = \{a, b\}$ ordered by the containment relation (non-exhaustive)

(?) preserves all internal relations of (?), despite the differences in the indexing. In particular, μ_1, μ_2, σ_1 in (?) project to μ_2, μ_3, σ_2 in (?), respectively.

The containment relation is particularly important because it provides a natural *ordering* over the space of possible structures. The ordering is not total, like a linear order; it is a *partial* ordering. In such a partially-ordered space, some structures are comparable in terms of containment, but not necessarily all of them are. Chandlee et al. (2019) provides some additional terminology: if one structure A is contained in another structure B , then structure A is a *subfactor* of structure B , whereas B is a *superfactor* of A . They prefer these terms to substructures and superstructures, which have different technical meanings in model theory.

To illustrate, Figure 4 displays a non-exhaustive portion of the partial ordering of possible strings over the alphabet $\Sigma = \{a, b\}$. At the bottom of the ordering is the empty string ϵ , with more complex strings appearing higher in the ordering. Whenever two elements are connected by a line, it means the higher element contains the lower element. For example, no line relates the a to b , because neither contains the other. However, a line is drawn from a to aa, ab and ba , because a is contained in each of these structures. As defined above, aa, ab, ba and aaa , and any other strings that contain a are superfactors of a ; correspondingly, a is a subfactor of each of these strings.

There can be many layers between a superfactor and one of its subfactors. When there are no structures ordered between a superfactor and its subfactor, it is called a *next superfactor*. In this sense, among all superfactors of a such as $\{aa, ab, ba, aaa, aab, \dots\}$, only those superfactors that are exactly one level above are next superfactors of a , namely $\{aa, ab, ba\}$.

Competing interests. The authors declare that there are no conflicts of interest regarding the publication of this paper.

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Data availability statement. A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings — e.g. Replication data and code can be found in Harvard Dataverse: <https://doi.org/link>.

Ethical standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

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Supplementary material. State whether any supplementary material intended for publication has been provided with the submission.

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